In [10]:
```
import numpy as np
X=np.array([
    [40,45],
    [42,43],
    [38,40],
    [75,78],
    [80,82],
    [78,76],
    [90,92],
    [88,89],
    [92,94]
])
```

In [15]:
```
from sklearn.cluster import KMeans
kmeans_model=KMeans(n_clusters=5, random_state=0)
kmeans_model.fit(x)
labels=kmeans_model.labels_
print(labels)
```

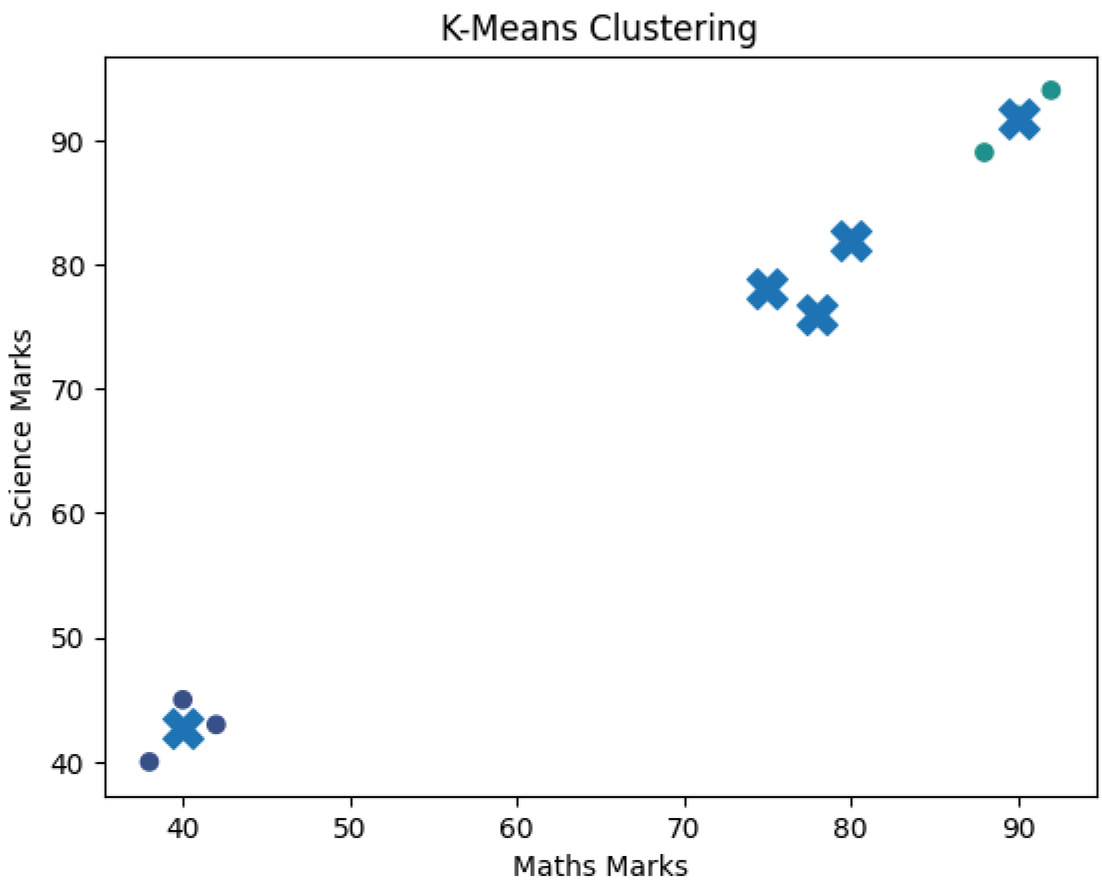
```
[1 1 1 3 0 4 2 2 2]
```

In [ ]:

In [16]:
```
import matplotlib.pyplot as plt

plt.scatter(X[:,0],X[:,1],c=labels)
plt.scatter(
    kmeans_model.cluster_centers_[:,0],
    kmeans_model.cluster_centers_[:,1],
    marker='X',
    s=200
)
plt.xlabel("Maths Marks")
plt.ylabel("Science Marks")
plt.title("K-Means Clustering")

plt.show()
```


K-Means Clustering
Science Marks
Maths Marks
90
80
70
60
50
40
40
50
60
70
80
90